PACMAN Zipline Firmware and Software (3.X)

Branches, builds, boards and status

Maintained by the PACMAN agents (augie)

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1 Purpose and scope

Zipline (3.X) is the current PACMAN firmware and software, for the 1v5 board (also built for the 1v4), after the legacy (1.X) and hijinks (2.X) generations (companion notes `legacy.tex`, `hijinks.tex`). This note records where zipline stands: the branches, what was built from which commit, which build is on which board, the results, and what is open. The firmware itself (registers, units, timing) is described in `firmware.tex`; the checkout tools in `qaqc.tex`. Digests are SHA-256, first eight hex digits.

2 Repositories

Repository	Branch	State
`mulhearn/pacman-firmware`	`working-on-3.6-zipline-2023.2`	928fcf0 on origin; firmware 3.6.0 candidate
`mulhearn/pacman-software`	`working-on-v3.6.0-zipline-2023.2`	4e9164c (290979d on origin)

- Toolchain: Vivado 2023.2 (firmware), PetaLinux 2023.2 (images and the Linux applications), Vitis 2023.2 (bare metal).
- Firmware simulation: GHDL, `ghdl/` (test benches) and `ghdl/agents` (`make check` against `docs/requirements.txt`); `vivado/src/hdl` is kept in step with `ghdl/` in each commit.
- Software: the Linux applications in `petalinux/pkg/pacman-apps` (`pacman_cmdserver`, `pacman_dataserver`, `pacman_menu`, `pacman_hwcheck`, `pacman_fwcheck`, `pacman_lnxcheck`), one copy of the drivers (`drivers-common`) shared with the bare-metal menu (`apps/menu`).

3 Firmware 3.6

3.6 adds to 3.5: the ADC unit, the logic analyzer with ADC capture and cross triggers, the `_UPPER`/`_LOWER` register names, the RX packet timeout in units of 256 cycles, and the RX sweep that resumes at the next channel after a packet closed by its maximum. It stays a candidate (3.6.0, not an official release) until it is released.

Commit	Date	Build	Result
650dea6	2026-10-26	hw1v5	bit 82c2be63, BOOT.BIN c43b1340; timing met (WNS +0.115 ns, WHS +0.049 ns); `pacman_fwcheck` PASS (30 of 30 requirements, twice)
ae71967	2026-10-29	—	on hw1v4, a reset on tiles 1–8 asserts the single legacy reset (was 1–10)
928fcf0	2026-10-30	hw1v4	bit 2755aa4c; timing met (WNS +0.185 ns, WHS +0.011 ns)

The hw1v5 bitstream of 650dea6 is the one in the current images: 928fcf0 changes only the hw1v4 build.

4 Software and images

Recent changes (pacman-software):

- 3d77617 to f516bc7 (2026-10-30): support for the ADC and logic analyzer; `pacman_hwcheck` as a test suite and `pacman_fwcheck`; the I2C overhaul (sticky status, bus recovery, MUX refusals as returns); one copy of the drivers; the I2C stress test.
- 2026-10-31: I2C at 100 kHz by default (640a124, needed on a hw1v4 card); ASIC TX at full drive in the configuration root (6d2ad8e); `--sma` tolerance 20% (41c6e00); the I2C device skip removed (3c9b2d0).
- 2026-11-01: the `pacman_hwcheck` tests as a submenu of both main menus (da113dc); one menu input on both platforms, a letter then Enter in every menu, a clean end at the end of input; one copy of the RX/TX and ADC menus for both platforms, the power menu folded into the I2C menu, a 16-BD ring for the menus (671a3dd); on the 1v4 since 2026-11-01.
- 2026-11-01 (290979d, which replaces 671a3dd with the changes below added): one RX/TX loopback core, which uses the configuration and ring as it finds them; in the RX/TX menu the ring size (16 to 512 BDs, default 512, set up by `m`), the loopback length, the RXTX UART mask (including tiles 2+3) and a BER check; `pacman_hwcheck --loopback` per tile, then all the masked tiles at once; the RX config and RX buffer config words built by functions. On the 1v4 since 2026-11-01.

The PetaLinux projects (`petalinux/ramdisk`, `petalinux/full`) are built in place, incrementally, after each software or firmware change:

Software	Date	Image	`BOOT.BIN`	`image.ub`
f516bc7	2026-10-30	ramdisk	41851385	7d452b37
3c9b2d0	2026-10-31	ramdisk	12813299	d7761858
3c9b2d0	2026-10-31	full	4c1d3a0e	8094cb12
671a3dd	2026-11-01	ramdisk	cec3182c	c4bd61b8
671a3dd	2026-11-01	full	96d3faca	8094cb12
290979d	2026-11-01	ramdisk	5bf59506	73368917
290979d	2026-11-01	full	348baf8a	8094cb12

5 Boards

Board	Firmware	Image	Results
1v5 (PACMAN 51 UCD)	650dea6	290979d (full SD image, root on p2)	2026-10-30, f516bc7: `pacman_hwcheck`, every test PASS on the tiles carrying the test cards, except `--ana_mon` on tile 4 (a weak TESTP leg, set aside). 2026-11-01, 290979d: `pacman_lnxcheck` 23 of 23 PASS (memory limits 700–800 MB, 4e9164c), `pacman_fwcheck` 30 of 30 PASS, `pacman_hwcheck` PASS on tiles 6+7 (loopback card), 9 (ATC card) and the board-wide tests; the production SD boot with TFTP fall-through PASS; LArPix v3 on tile 1 replies and passes the ASIC loopback (`larpix.tex`). Bare-metal menu (4e9164c): boot, versions, I2C, hwcheck read-only, ADC, I2C stress and the ASIC hello PASS; the RX/TX and ASIC loopbacks lose words (open)
1v4	928fcf0	290979d (TFTP)	2026-11-01: the new menus work (letters and Enter, a clean end at the end of input, the hwcheck submenu: read only as expected, I2C stress PASS); the LArPix v3 on tile 1 does not reply (UART 0 streams junk; it replied exactly on 2026-10-31 with 3c9b2d0): the configuration sent right after the full reset is lost, and a 0.2 s pause fixes it. 2026-11-01, 290979d: a 16-BD ring set up from the menu, the ASIC hello exact twice, and the BER loopback on tiles 2+3

The 1v4 has no SD card: it boots by JTAG and TFTP; it is off the bench since 2026-11-01. On the 1v5, ssh and scp to the board work by key (sshd allows root by key only).

6 Open

- Release of 3.6 (official).
- The ramdisk image boots with its ramdisk inside the 256 MB DMA buffer (U-Boot places it at 2b2ee000–2ffff292), so the buffer's reservation fails and the kernel uses that memory; fix: `initrd_high` and `fdt_high` below 0x20000000. The full SD image is not affected.
- Bare-metal RX/TX loopback: 79255 of 80000 words received, nearly all miscompared, identical on fresh starts and with the data cache off; Linux runs the same code without error.
- Robustness of the Vivado build: `create.tcl` adds the HDL from the repository, so HDL edited after the project was created partly enters a rebuild.
- Set aside: the `pattern.vhd` start-up (one write that enables and moves the phase can cut the first start bit short), the DMA race (the dataserver and `hwcheck`/menu share the RX ring: stop the servers first), private rootfs customizations, `pacman_setup.sh` at first boot, an I2C termination option, an I2C error state in a register, a richer I2C simulation.